

Virtual Reality-Based Interactive System for Enhanced Real Estate Exploration and Custom Interior Design

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Abstract

Traditional real estate visualization methods like 2D drawings and static images often fail to help clients fully understand space and design potential. Virtual Reality (VR) addresses this by offering immersive, interactive experiences where users can explore and customize virtual properties in real time. This study introduces a VR system built with Unity 3D, XR Toolkit, and Blender, featuring 3D walkthroughs, real-time color changes, dynamic lighting and weather, and drag-and-drop furniture adjustments. The development followed a clear process: analyzing user needs, creating virtual environments, integrating interactive features, and thorough testing on VR headsets. Performance tests showed smooth, high-quality visualization and intuitive interaction. This platform empowers architects, real estate agents, and clients to communicate design ideas effectively and make quicker, more informed decisions. By combining immersive visualization with active customization, the system enhances presence and supports sustainable design choices. It also aligns with current trends in VR use for real estate, which highlight improved buyer engagement, reduced decision times, and broader market reach. Future improvements may include AI-based design suggestions and collaborative features, expanding VR's role from a visualization tool to a comprehensive decision-making solution.

Keywords

Virtual Reality, Real Estate, Interior Design, Immersion, Customization

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1 Introduction

The real estate and interior design industries rely heavily on visual communication to convey spatial qualities, aesthetics, and design concepts. Traditionally, this has been achieved through 2D drawings, static 3D renderings, or physical models. While effective to

some degree, these methods often fail to provide clients with a true sense of space, scale, and customization. As a result, decision-making can be slow and uncertain, with clients struggling to imagine how a property will look or feel once completed. Virtual Reality (VR) has emerged as a promising solution to these limitations. By creating immersive, interactive, and realistic environments, VR allows users to explore spaces as if they were physically present. In real estate marketing, VR tours have been shown to improve customer attitudes and influence purchase intentions by evoking stronger emotions and a greater sense of presence compared to static imagery [4, 5, 16]. In interior design, VR applications enable real-time customization of colors, furniture layouts, and environmental conditions, offering both efficiency and a higher degree of user satisfaction [9, 11]. The relevance of VR extends beyond visualization alone. Recent advances highlight its role in interactive design, where users can directly manipulate spatial elements, and in generative and mixed reality approaches, where AI and MR technologies simplify workflows and expand creative possibilities [13, 14]. At the same time, VR contributes to broader digital transformation within the built environment, supporting sustainable design, energy-efficient planning, and user-centered architectural practices [15]. Despite these advances, challenges remain. VR design workflows can be hardware-intensive, the learning curve for first-time users can be steep, and large-scale adoption in real estate and design industries is still in its early stages. These gaps present opportunities for developing more efficient, intuitive, and accessible VR systems tailored to the needs of architects, designers, real estate agents, and clients. This research aims to develop and evaluate a VR-based interactive system that enhances real estate exploration and interior design customization. Specifically, the system enables immersive 3D walkthroughs of residential spaces, real-time adjustments of wall and furniture colors, drag-and-drop furniture manipulation, and environmental controls such as lighting and weather conditions. By combining these features into a unified platform, this study seeks to bridge the gap between architectural design and client experience, supporting faster, more informed, and more satisfying decision-making.

2 Related Work

2.1 VR in Real Estate Marketing and Visualization

Virtual reality has become a powerful tool in real estate marketing, especially for property visualization. Early studies suggested that non-immersive VR, such as 360° tours on websites, improves customer experience and attitudes compared to static images [16]. Later research confirmed that immersive VR tours not only create

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stronger feelings of presence but also influence purchase intentions by shaping emotions like pleasure and arousal [5, 13]. More recent findings show that even low-immersion VR can support decision-making and increase sales effectiveness, although head-mounted displays (HMDs) provide deeper emotional engagement [3]. These works highlight VR's role in reducing uncertainty for buyers and building trust in property marketing.

2.2 VR for Interior Design and Customization

Alongside marketing, VR has gained traction as a design tool in interior design. One study demonstrated that VR-based design platforms, when optimized with Level of Detail (LOD) techniques, significantly improve rendering efficiency and enhance the visual quality of interior spaces [14]. Similarly, another investigation found that while VR design tasks can take longer than desktop-based design, users report higher satisfaction and stronger spatial understanding in immersive environments [6]. A green interior design framework has also been proposed using VR to promote sustainability and reduce environmental impact [2], while augmented VR systems that combine panoramic imagery with 3D interaction have shown promise for remodeling decisions [10]. Collectively, these studies show that VR not only improves efficiency and user satisfaction but also broadens the scope of design to include sustainable practices and hybrid visualizations.

2.3 Interaction and Generative Approaches in VR/MR

Recent work has explored how VR can move beyond visualization to become a platform for active interaction. A generative AI-driven Mixed Reality (MR) method was developed to support real-time spatial modification, enhancing design flexibility and interactivity [9]. This approach shows that combining MR with generative algorithms simplifies traditional workflows and improves user engagement. Similarly, research in the furniture, fixture, and equipment sector highlighted how distributed and single-user VR improves design communication, but also faces adoption barriers due to cost [11]. Together, these studies illustrate a shift toward VR/MR as collaborative and generative environments where users can directly shape space and design.

2.4 Presence, Usability, and User Experience in VR Environments

A recurring theme in the literature is how VR enhances presence and user experience in architectural and design contexts. It has been emphasized that improving presence requires careful design of real-time visualization content [8]. Empirical studies show that immersive VR environments elicit stronger emotional responses, higher physiological arousal, and more positive attitudes compared to non-immersive experiences [3]. However, challenges remain: VR design can be more time-consuming, and usability depends on hardware performance and user familiarity [3, 7]. Despite these challenges, VR consistently demonstrates value in making design more intuitive and engaging for clients and professionals.

2.5 Digital Transformation and Sustainable Design

Beyond individual projects, VR is increasingly linked to broader digital transformation in the built environment. A review of research trends called for adaptive, user-centered VR systems for architecture and construction [11]. A more recent study extended this discussion by reviewing how digital technologies such as VR, AR, AI, and digital twins are reshaping real estate operations, highlighting benefits but also challenges like cost and integration barriers [15]. Meanwhile, VR has been applied in low-carbon interior design, showing how immersive environments can support energy-efficient design decisions [12]. These works suggest that VR is not only a visualization tool but also a driver of sustainable and digital innovation in the real estate sector.

3 Methodology

The development of the proposed VR-based system followed a structured process that combined technical design with iterative testing. The overall aim was to build an immersive environment where users could not only explore a property but also interact with its interior features in real time.

3.1 Technology Stack

The VR-based interactive system was built using a comprehensive technology stack designed to ensure optimal performance, cross-platform compatibility, and seamless user interaction. The following technologies were integrated to create a robust and scalable solution:

- **Unity 3D** serves as the primary development platform and game engine. Unity was selected for its extensive VR support, real-time rendering capabilities, and cross-platform deployment features. Its built-in physics engine, lighting system, and asset pipeline provided essential foundations for creating immersive 3D environments. Unity's real-time rendering allows for dynamic visualization where lighting, shadows, and textures update instantly as users navigate through virtual spaces.
- **C#** functions as the primary programming language for system logic and functionality implementation. All core interactive features, including object manipulation, user interface controllers, and system state management, were developed using C#. The language's object-oriented architecture facilitated modular development and efficient memory management, crucial for VR applications.
- **JavaScript** handles real-time color picker functionality, dynamic UI updates, and environmental control systems. Its event-driven programming model enables responsive user interactions with minimal latency, essential for maintaining immersion in VR environments.
- **Blender** served as the 3D modeling and asset creation tool for all architectural elements, furniture models, and environmental assets. Blender's comprehensive modeling tools, UV mapping capabilities, and texture painting features enabled creation of optimized 3D assets. Models were designed with careful attention to polygon count and texture resolution to

ensure smooth real-time performance while maintaining visual fidelity. The software's export compatibility with Unity's asset pipeline streamlined the development workflow.

- **VR headsets** enable users to observe and navigate 3D spaces naturally, movement tracking for head position and orientation. This allows users to lean, crouch, and move around virtual furniture and architectural elements as they would in physical spaces. The headsets' high refresh rates minimize motion sickness while maintaining immersion during extended exploration sessions. VR Headset: Oculus Quest 2 (128 GB) – used for immersive testing and real-time interaction. Intel Core i7 (12th Generation) – ensured high-speed computation for rendering and input processing. NVIDIA GeForce RTX 3070 (8 GB GDDR6) – provided smooth 90 FPS+ frame rates and realistic lighting performance. 8 GB DDR4 – supported stable multitasking during Unity builds and Blender asset integration. Windows 11 64-bit – platform for Unity 3D, Blender, and XR Toolkit integration.

This technology combination provides a balanced architecture that prioritizes both visual quality and performance optimization, essential for delivering smooth VR experiences across different hardware configurations.

3.2 Workflow

The process began with a requirement analysis, during which user needs were collected from architects, interior designers, and potential clients. This informed the design of sketches and floor layouts that served as a blueprint for the virtual environment. In the environment creation phase, 3D assets were modeled in Blender and imported into Unity. Lighting was carefully configured to simulate both natural and artificial sources, forming the basis for immersive visualization. Next, interactive elements were integrated. A real-time color picker allowed changes to wall and furniture colors, while a drag-and-drop function enabled furniture rearrangement. Environmental features such as day/night cycles and weather conditions (sunny, cloudy, rainy) were implemented to expand the realism of the experience, figure-1. Finally, the system underwent testing and iteration. Prototype builds were deployed on VR headsets, and performance was evaluated in terms of rendering smoothness, responsiveness of interaction, and overall usability. Issues such as lag and navigation difficulty were addressed through debugging and optimization until a stable version was achieved.

3.3 Workflow Summary

The overall workflow is summarized in Figure 1, which outlines the four key phases of the development process. It begins with requirement analysis, where user needs and design goals are identified, followed by environment creation using 3D modeling and lighting setup. The third phase focuses on integrating interactive features such as color customization, furniture manipulation, and environmental controls. Finally, the system undergoes testing and iterative refinement to ensure stability, usability, and immersive performance.

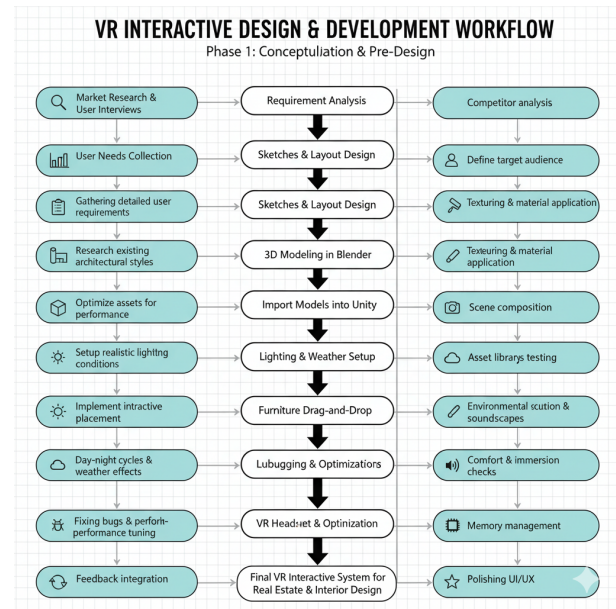


Figure 1: Workflow of the VR-based interactive system

4 System Features

The VR-based interactive system was designed as a multifunctional platform that addresses key limitations of traditional design and visualization methods. Instead of relying solely on static 3D renders or floor plans, the system enables clients and professionals to explore and interact with a space dynamically. This section presents the major features of the system and discusses their significance. The most fundamental capability is the immersive 3D walkthrough. Users can navigate through a virtual residential property at full scale, experiencing spaces as though they were physically present. This feature is crucial because one of the major challenges in real estate and interior design lies in helping clients understand scale, proportions, and circulation. In traditional 2D floor plans, clients often struggle to interpret spatial relationships, while static 3D renderings present only a fixed perspective. By contrast, the walkthrough provides freedom of movement and exploration, allowing users to judge room sizes, ceiling heights, and furniture placement naturally and intuitively, figure-2.



Figure 2: Walkthrough mode inside the living room environment.

Another core feature is real-time color customization. The system allows users to select walls, floors, and furniture surfaces and instantly apply alternative colors, figure-3. This functionality eliminates the need to generate multiple renderings for each design option, which is often time-consuming in traditional workflows. Instead, clients can experiment directly with different design palettes in real time, comparing how various color schemes affect the atmosphere of the space. Beyond aesthetic flexibility, this feature also enhances communication between designers and clients by providing immediate feedback during consultation sessions.



Figure 3: Wall color being modified through the integrated color picker.

The third feature is dynamic environmental control, which enables users to alter lighting and weather conditions within the virtual property. This includes switching between daytime and nighttime modes, simulating artificial lighting scenarios, and applying different outdoor conditions such as sunny, cloudy, or rainy weather, figure-4. This function is especially valuable in architectural and interior visualization, where the appearance of a room can change dramatically depending on natural light exposure or artificial lighting design. By providing these variations, the system allows clients to make more informed decisions about window placement, lighting fixtures, and material finishes.

Finally, the platform incorporates interactive furniture manipulation. Users can select and reposition furniture items through a drag-and-drop interface, testing different layouts to assess comfort and functionality, figure-5. This feature turns the system into more than a presentation tool—it becomes a collaborative design environment. Clients are no longer passive observers; they can participate actively by arranging spaces according to their preferences, while designers can demonstrate alternative layouts instantly. This interactivity supports user-centered design and strengthens client confidence in decision-making.

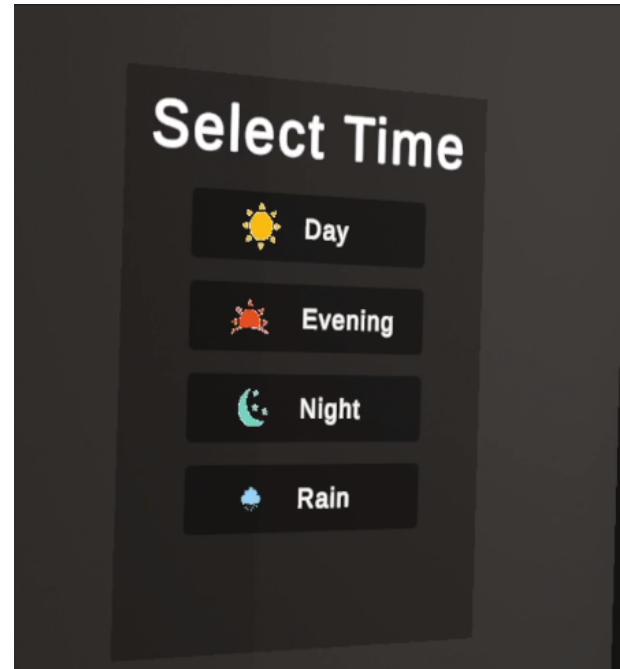


Figure 4: Menu to view same space under daylight and evening conditions.



Figure 5: A furniture rearrangement scenario in the living area.

5 Results and Evaluation

The evaluation of the VR-based interactive system focused on three dimensions: technical performance, usability, and user experience. Testing was conducted using VR headsets to replicate real usage conditions, and the results are compared with findings from related literature to position the contribution of this work.

5.1 Performance and Rendering

Smooth performance is critical for immersion, as lag or frame drops can break the illusion of presence and cause discomfort. In our evaluation, the system consistently maintained high frame rates, averaging above 90 FPS during walkthroughs. Rendering optimizations—such as applying Level of Detail (LOD) techniques for complex models and pre-baked lighting for static elements—contributed

significantly to reducing rendering times. This allowed real-time customization features, such as color changes and weather transitions, to be executed without noticeable delay. These findings are consistent with previous work showing that optimized VR platforms can enhance both speed and visual fidelity [2, 15].

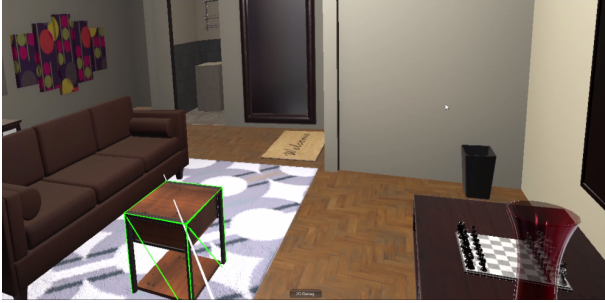


Figure 6: A stable rendering scene during a walkthrough, demonstrating smooth visual performance.

5.2 Usability and Interactivity

The usability of the system was assessed through exploratory sessions in which participants interacted with its features. Navigation was generally intuitive, with users adapting quickly to VR headset controls. The drag-and-drop furniture interaction was described as both engaging and practical, while the real-time color customization was highlighted as a feature that simplified design discussions. The environmental controls were also well received, as they allowed clients to visualize properties under conditions that closely matched real-world variability. These results mirror findings in recent studies, where VR systems were shown to improve design communication and client satisfaction compared to traditional tools [4, 5, 7].

5.3 Immersion and User Experience

Beyond functionality, the system's ability to generate immersion was a key strength. Participants reported a strong sense of presence, particularly when moving through rooms and observing spaces at life scale. The ability to alter conditions—such as changing daylight to nighttime—contributed to this immersive effect by giving users control over environmental dynamics. Compared with conventional visualization methods, the VR walkthrough produced a more emotional and engaging experience, aligning with previous findings that immersion improves client decision-making in real estate [11, 16].

5.4 Limitations and Future Improvements

Despite these positive results, several limitations were observed. The system currently depends on high-performance hardware to maintain smooth interaction, which may restrict accessibility for clients without advanced devices. A short learning curve was also noted for first-time VR users, though this is consistent with broader trends in VR adoption [1, 14, 15]. Additionally, while the current features provide strong interactivity, more advanced capabilities—such

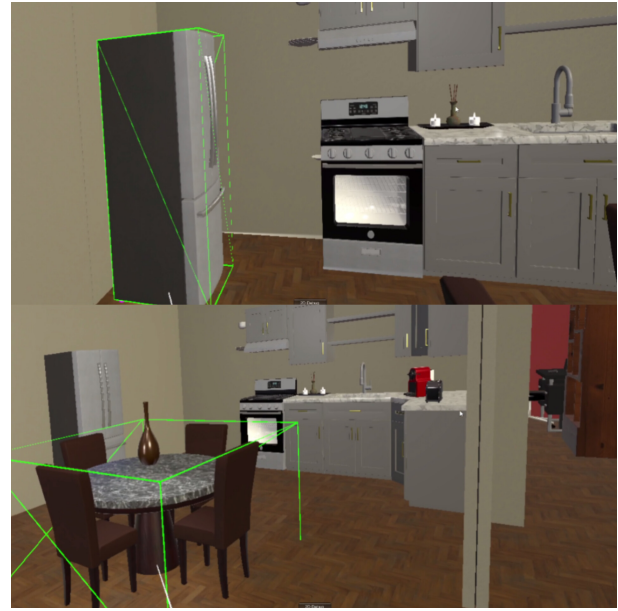


Figure 7: A user rearranging furniture inside the living room environment.



Figure 8: The same room under day and evening lighting, highlighting the immersive impact of environmental variation.

as AI-driven design suggestions or collaborative multi-user environments—remain unexplored. Future development should address these gaps to further increase system usability and scalability.

6 Discussion

The developed VR system demonstrates clear advantages over traditional visualization methods by providing immersive walkthroughs and interactive customization features. Unlike static drawings and 2D plans, the platform allows clients to actively engage with design options, which enhances understanding and confidence in decision-making. This aligns with earlier findings that VR increases presence and supports stronger emotional engagement in real estate marketing.

A key contribution of this study is the integration of multiple features into one platform. Previous works have typically focused on isolated aspects, such as walkthroughs or color customization [4, 15, 16]. By combining furniture manipulation, lighting and weather simulation, and dynamic walkthroughs, our system creates a more holistic and flexible environment for both real estate exploration and design experimentation. This supports earlier calls for adaptive and user-centered VR solutions [11].

The study also contributes to discussions on user experience. While immersive VR improves satisfaction and realism [4, 9], challenges such as hardware demands and learning curves remain [10, 12]. These limitations echo similar concerns raised in studies of architectural visualization [10], VR adoption in the furniture and equipment sector [1], and even hospitality management, where usability and cost are critical barriers [2].

The broader literature suggests that digital transformation in real estate is accelerating through technologies such as VR, AR, AI, and digital twins [8]. Recent reviews have shown that these technologies bring efficiency but also face barriers in cost, data security, and integration [4, 6]. Our findings reinforce this perspective: while VR offers clear benefits, widespread adoption will require lowering technical barriers and streamlining workflows.

Finally, VR's role in sustainability and green design deserves emphasis. Research has shown how VR supports low-carbon and environmentally responsible design decisions [7, 9]. Pandemic-driven studies also highlighted VR as a resilient marketing tool when physical interactions were restricted [17]. Similarly, distributed and remote visualization approaches support design communication across industries, from real estate to interior design and hospitality [16]. These directions suggest that VR is not just a visualization medium but an evolving decision-support system that will continue to shape how spaces are designed, marketed, and experienced.

7 Conclusion and Future Work

This study developed a VR-based interactive system for real estate exploration and interior design customization. The platform combines immersive walkthroughs with features such as real-time color adjustments, furniture manipulation, and dynamic lighting and weather control. By offering an integrated and interactive environment, the system addresses the limitations of traditional 2D drawings and static renders, helping clients better understand and engage with design options. The results demonstrate that VR enhances presence, user engagement, and decision-making efficiency, while supporting clearer communication among architects, designers, and real estate professionals. However, challenges such as hardware demands and the learning curve for new users remain. Future work should focus on expanding accessibility, integrating AI-driven

design recommendations, and enabling multi-user collaboration. With these improvements, VR has the potential to evolve from a visualization tool into a comprehensive decision-support system for the built environment.

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